Predicting nominal charge from EEG

Tasks 3 and 4 | RC-007 and RC-008 Satyam Awasthi 13 November 2026

Post-stimulus EEG gave no reliable improvement over the training-fold mean charge. Task 3 varied charge within each waveform and provided the direct test of dose prediction. Task 4 used one nominal charge per waveform within a participant and tested how well a voltage-only model predicted charge across comparisons and waveform types.

Shared model and controls

Target and predictors

The target was nominal cathodic charge, in μC, calculated as

$$Q_{\text{nom}} = I_{\text{peak}} \times 0.4\,\text{ms} \times A_{\text{shape}},$$

where I_{peak} is the recorded peak-current setting in mA and A_{shape} is the dimensionless area factor for the ideal waveform. Task 3 supplied the trial amplitude setting directly. Task 4 used the participant-by-waveform amplitude lookup inferred from the CSV's second presentations in WD-007. These values describe the programmed stimuli. Delivered charge would require the current-monitor recordings.

The regressions used 126 voltage features per trial, with nine consecutive 50 ms means at each of 14 scalp electrodes. The post-stimulus window was 50 to 500 ms. The matched pre-stimulus window was -500 to -50 ms. Both used scalp-average referencing and their own earlier voltage baseline.

Fitting and evaluation

For each participant and EEG window, ridge regression fit $\widehat{Q} = b_0 + \mathbf{b}^\top \mathbf{x}$ using EEG voltage as the sole predictor. An inner validation loop selected the ridge penalty from $1, 10, 100, 1{,}000, 10{,}000$ by mean squared error. Feature standardization was fitted inside each training split. Pre- and post-stimulus models used identical outer test trials. The pre-stimulus model measures how much prediction is possible before the current pulse.

How to read the scores

Each outer fold had a constant reference that predicted the mean nominal charge of its training trials. Mean absolute error (MAE) gives the typical prediction error in μC. Relative $R^2 = 1 - \text{MSE}_{\text{EEG}}/\text{MSE}_{\text{mean}}$ measures the reduction in squared error versus that training-fold mean. A negative value means the EEG model had higher squared error. Scores weight participants equally. Task 3 also weights waveforms equally within participant. The 95% intervals resample participants, pairing the two EEG windows for pre/post comparisons.

Task 3 | Charge varies within each waveform

The reviewed alignment included 1,469 trials from 10 participants and all 50 participant–waveform groups. The original strict alignment retained 1,382 trials in 47 groups. The reviewed join recovered three groups using verified CSV blocks and the exact trigger sequence after applying the documented exceptions.

Within-waveform test. Each fit used one participant and one waveform. The outer folds held out chronological trial chunks, with all three amplitude tiers present in every outer training fold. Inner tuning also held out chronological chunks. This tests whether voltage predicts dose among trials of the same waveform.

Held-out-waveform test. Each fit used the remaining waveforms of one participant and predicted charge for trials of the entirely held-out waveform. Inner tuning also held out whole waveforms. Other waveforms can share the same charge.

Reviewed alignment results

Held-out unit	EEG window	EEG MAE (μC)	Mean MAE (μC)	Relative R^2 [95% interval]
Within-waveform chunk	Pre	0.04718	0.04527	−0.096 [−0.167, −0.040]
Within-waveform chunk	Post	0.04642	0.04527	−0.032 [−0.084, +0.018]
Entire waveform	Pre	0.08226	0.08176	−0.013 [−0.024, −0.002]
Entire waveform	Post	0.08035	0.08176	+0.022 [−0.026, +0.087]

Post minus pre MAE was −0.00076 μC within waveform (95% interval −0.00185 to +0.00034) and −0.00191 μC for held-out waveforms (−0.00448 to +0.00007).

Charge coverage when a waveform was held out

Target charge relative to training	Trials	Mean MAE (μC)	Pre MAE (μC)	Post MAE (μC)
Seen in a training waveform	920	0.06059	0.06048	0.06031
New, within training range	331	0.06667	0.06747	0.06685
Outside training range	218	0.18551	0.18499	0.17794

The small overall post-stimulus gain in the transfer test did not extend to the 331 trials with new charges inside the training range. The larger absolute errors outside that range reflect the harder extrapolation test.

Strict-alignment sensitivity

In the strict run, post-stimulus MAE within waveform was 0.04694 μC versus 0.04583 μC for the mean, with relative $R^2 = -0.031$. For held-out waveforms, it was 0.08250 μC versus 0.08332 μC, with relative $R^2 = +0.008$ and a 95% interval of −0.040 to +0.075.

Task 4 | One nominal charge per waveform

The analysis used 1,244 trials from 10 participants. Within each participant, nominal charge is fixed for a given waveform, so a model could exploit a waveform-related EEG pattern when predicting charge.

Comparison holdout. The five original WD-001 folds held out complete pairwise waveform comparisons and kept both presentations together. Every test trial had a nominal charge already present in training.

Waveform holdout. Each fold removed from training all comparisons involving the test waveform, including its paired presentations. The model was scored on trials of the held-out waveform. Inner validation in both Task 4 tests grouped complete comparisons. Training excluded pairs whose EEG support could overlap held-out EEG under the 1.25-second purge rule.

Held-out results

Held-out unit	EEG window	EEG MAE (μC)	Mean MAE (μC)	Relative R^2 [95% interval]
Pairwise comparison	Pre	0.04885	0.04834	−0.075 [−0.209, −0.003]
Pairwise comparison	Post	0.04858	0.04834	−0.017 [−0.051, +0.012]
Entire waveform	Pre	0.05975	0.05812	−0.047 [−0.082, −0.019]
Entire waveform	Post	0.05900	0.05812	−0.018 [−0.031, −0.005]

Post minus pre MAE was −0.00027 μC for held-out comparisons (95% interval −0.00098 to +0.00040) and −0.00075 μC for held-out waveforms (−0.00219 to +0.00039).

Charge coverage when a waveform was held out

Target charge	Trials	Participants	Mean MAE (μC)	Pre MAE (μC)	Post MAE (μC)
Seen in training	543	8	0.03286	0.03335	0.03413
New, within range	315	8	0.04103	0.04509	0.04228
Outside range	386	10	0.10483	0.10528	0.10422

Each row averages participants with eligible trials. Post-stimulus EEG gave higher MAE than the training mean for seen charges and new in-range charges.

What the two tasks establish

The within-waveform Task 3 test and both Task 4 tests gave higher post-stimulus MAE than the training mean. Task 3 waveform transfer showed a small and uncertain gain. Across both tasks, the paired pre/post intervals included zero. The evidence is limited to single-trial predictions using these voltage features and a linear model.

Analysis records in `Docs/RC008_TASK3_CHARGE_PREDICTION.md` and `Docs/RC007_TASK4_CHARGE_PREDICTION.md`.